

Solving systems of polynomial equations via Macaulay matrices

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Abstract. One approach to solving polynomial systems is to multiply each equation by monomials, which creates a larger system with the coefficient matrix known as the Macaulay matrix. The eXtended Linearization (XL) method, introduced by Courtois, Klimov, Patarin, and Shamir in 2000, is one such approach and includes a sub-algorithm that performs Gaussian elimination on the Macaulay matrix. Due to the simplicity of the method, several improvements and variations have been proposed since its introduction, and it remains an active area of research. In this paper, we focus on sub-algorithms based on Macaulay matrices that appear in the XL method and its variants and investigate input parameters that yield desired outputs such as Gröbner bases. In particular, by revisiting the standard degree setting and formulating a framework without fixing a specific monomial order, we provide a foundation for extending the analysis to the multi-degree case.

Keywords: solving polynomial systems · Macaulay matrix · Gröbner basis · XL algorithm · multi-degree

1 Introduction

The problem of solving systems of polynomial equations is a fundamental topic in mathematics and arises naturally in various fields, including coding theory and cryptography. In particular, the problem of finding solutions over finite fields plays a crucial role in these areas. It is known that this problem is NP-hard even when the defining polynomials are quadratic over a binary field. Moreover, various algorithms have been actively studied to better understand its complexity.

A common approach to solving a system of polynomial equations is to construct an extended system by multiplying each defining equation by suitable monomials. The coefficient matrix of the resulting system is known as the Macaulay matrix, which has been studied in the context of resultants since at least [22]. For instance, Lazard [19,20,21] proposed an algorithm that utilizes an echelon form of the Macaulay matrix to solve polynomial systems.

The eXtended Linearization (XL) method [7] is an algorithm for solving polynomial systems formalized by Courtois, Klimov, Patarin, and Shamir in 2000 for large-scale polynomial systems, and it also utilizes an echelon form of

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the Macaulay matrix. Due to its simplicity, several improvements and variations of the method have been proposed since its introduction [28,29,11], including algorithms for solving polynomial systems based on the computation of kernel vectors of the Macaulay matrix, such as [29]. It remains an active topic of research.

Given a polynomial system F and an input parameter $D \in \mathbb{Z}_{\geq 0}$, the XL method generates many polynomial equations by multiplying each defining polynomial f by monomials of degree at most $D - \deg f$ and computes an echelon form of the Macaulay matrix, denoted by $Mac_D(F)$, of the resulting system. To find a solution, the XL method in [7] requires that this echelon form contains a univariate polynomial. By solving this polynomial, it reduces to a polynomial system with fewer variables, and a solution is obtained by repeating this process. In this context, it is known that, for a given polynomial system F , the input degrees D for which the XL algorithm produces a univariate polynomial have been characterized in works such as [23,9], providing a theoretical understanding of when such elimination occurs.

The XL method seeks a univariate polynomial contained in the ideal generated by the input polynomial system and is fundamentally based on elimination theory. In fact, if the ideal is zero-dimensional, the Gröbner basis with respect to the lexicographic monomial order contains a univariate polynomial. Several algorithms based on XL have been developed, and their objective is not necessarily to obtain a univariate polynomial in the echelon form. For example, if we have a Gröbner basis with respect to a certain monomial order, we can compute a Gröbner basis with respect to the lexicographic order using a monomial order conversion algorithm such as FGLM [13]. Moreover, analyses that characterize input degrees for which Gröbner bases with respect to the graded reverse lexicographic order are computed have been presented in [6,16].

1.1 Our contribution

Motivated by this background, we study the linear-algebra step of XL-type algorithms through the Macaulay matrix and investigate the role of the input degree D in determining both the outcome and the complexity of this step. In particular, we provide a systematic framework, based on generating functions associated with the Macaulay matrix, that does not depend on the choice of monomial order, which allows one to predict when solution-relevant relations or Gröbner bases appear and to derive corresponding complexity estimates. From a conceptual viewpoint, this provides a formulation of the linear-algebra steps underlying XL-type methods, including both echelon-based and kernel-based approaches, and shows that their complexity analysis can be reduced to the determination of the associated generating series.

At a high level, we characterize admissible input degrees by analyzing the dimension growth of certain subspaces associated with the Macaulay matrix. Firstly, regarding the ideal $\langle F \rangle$ generated by the defining system as a vector space over $\mathbb{F}$, we introduce a subspace, denoted $\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}$, which is a subspace of the space of polynomials of degree at most D , denoted by $A_{\leq D}$, and is isomorphic to

the column space of the Macaulay matrix $Mac_D(F)$, together with a generating function of the sequence $\dim_{\mathbb{F}} A_{\leq D} / \langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}$. We provide characterizations using this generating function to various admissible input parameters D for which the sub-algorithm performing Gaussian elimination on Macaulay matrices returns a univariate polynomial, a linear polynomial, or a Gröbner basis. The smallest admissible input degree that yields the desired output is often referred to as the *operating degree* (see, e.g., [29]), and our results provide upper bounds for this quantity.

For instance, when the system has a unique solution, any degree $D \geq 1$ such that $\dim_{\mathbb{F}} A_{\leq D} / \langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}} = 1$ — that is, the number of columns minus the rank is one — guarantees that the sub-algorithm outputs a Gröbner basis with respect to any fixed monomial order. In particular, we show when the generating function coincides with

$$\left[\frac{(1-t^2)^m}{(1-t)^{n+1}} \right]_{(1)}; \text{ (see Definition 9),}$$

we can combinatorially determine admissible input parameters D that yield a Gröbner basis.

These discussions depend on the existence of a degree D such that $\dim_{\mathbb{F}} A_{\leq D'} / \langle [A \cdot F]_{\leq D'} \rangle_{\mathbb{F}} = 1$ for any $D' \geq D$, and it is characterized by the condition $\dim_{\mathbb{F}} A / \langle F^{top} \rangle < \infty$, where F^{top} is the system consisting of the top homogeneous components of the polynomials in F .

For systems of polynomial equations with a multi-degree, the condition $\dim_{\mathbb{F}} A / \langle F^{top} \rangle < \infty$ does not necessarily hold, and systems that do not satisfy this condition can be easily found. For instance, for consistent bilinear systems, this condition may fail, yet the sub-algorithm can still compute a Gröbner basis for a certain input degree.

To illustrate this, we investigate subspaces $\langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}}$ with $\mathbf{D} \in \mathbb{Z}_{\geq 0}^s$ in the multi-degree setting, together with an associated generating function of the dimension sequence of these quotient spaces, and a multi-degree version of the sub-algorithm, where the monomials used for extending the system are restricted by multi-degree. For example, if $\mathbf{D} \in \mathbb{Z}_{\geq 1}^s$ satisfies $\dim_{\mathbb{F}} A_{\leq \mathbf{D}} / \langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}} = 1$, then the multi-degree sub-algorithm returns a Gröbner basis of the ideal $\langle F \rangle$.

Moreover, we also study a second sub-algorithm on the same Macaulay matrix, namely the computation of right kernel vectors, and investigate its multi-degree version. In this approach, since a non-trivial kernel vector always exists when the system is consistent, we suggest estimating its complexity at each intermediate degree.

As a result, we show that, in actual cryptanalysis [1,5,25], the sub-algorithm included in XL and its variants computes a Gröbner basis. Furthermore, together with the kernel vector computation approach, we provide upper bounds on the complexity, taking into account each intermediate degree.

Our results build on the detailed discussion and experimental results provided in [1,25,28,29]. This paper is not intended to propose practical algorithms, nor to provide new complexity estimates. Rather, while the works [1,5,25] focus on

the analysis of specific polynomial systems arising from concrete cryptanalytic constructions, our contribution is to provide a general framework.

1.2 Related works

- In this paper, we focus on a sub-algorithm of the XL algorithm introduced by Courtois et al. [7]. This perspective is fundamentally different from studies on iterative algorithms such as Mutant XL [11], F4 [12], and F5 [14], which analyze the behavior of the entire algorithm. In contrast, we focus on a single linear-algebraic step at a fixed degree and analyze its complexity independently of any iterative process.
- Consider a quadratic system that has a unique solution in affine space over the algebraic closure and no solutions at infinity. In this case, the smallest input degree among those characterized in this paper ensures that a Gröbner basis with respect to any monomial order can be computed. In particular, this value serves as an upper bound for the operating degree (see, e.g., [29]). Moreover, it is smaller than or equal to the values given in [6] and the proof of Theorem 30 in [16] for the graded reverse lexicographic (grevlex) order (see Remark 8). Unlike [6,16], whose analysis relies on the grevlex order, our analysis does not fix a specific monomial order and is independent of the choice of order. Further related results are discussed in Remark 8.

1.3 Organization of the paper

The paper is organized as follows. In Section 2, we recall basic concepts and a sub-algorithm used in XL and its variants. In Section 3, we investigate the input parameters under which sub-algorithms produce the desired outputs in the standard degree case. In Section 4, we extend the discussion in Section 3 to the multi-degree case. In Section 5, we restate the results from Sections 3 and 4 for input polynomial systems with a certain combinatorial property and provide an upper bound on the complexity of the sub-algorithm.

2 Preliminaries

2.1 Notation

In this paper, vectors are written as row vectors.

Let $\mathbb{F}$ be a field and $A = \mathbb{F}[x_1, \dots, x_n]$ be the polynomial ring in n variables $x_1, \dots, x_n$ over the field $\mathbb{F}$. For a set $F \subseteq A$, we denote by $\langle F \rangle_{\mathbb{F}}$ the vector space spanned by F and by $\langle F \rangle_A$, or simply $\langle F \rangle$, the ideal generated by F :

$$\langle F \rangle_{\mathbb{F}} := \{c_1 f_1 + \dots + c_m f_m \mid c_i \in \mathbb{F}, f_i \in F\},$$

$$\langle F \rangle_A := \{g_1 f_1 + \dots + g_m f_m \mid g_i \in A, f_i \in F\}.$$

Note that A and $\langle F \rangle_A$ are regarded as a vector space over $\mathbb{F}$. For the set $\underline{X} = \{x_1, \dots, x_n\}$ of variables, we denote by $Mon(\underline{X})$ the set of all monomials in A :

$$Mon(\underline{X}) := \{x_1^{e_1} \cdots x_n^{e_n} \mid e_i \geq 0\}.$$

Moreover, we denote by $Mon_d(\underline{X})$ the set of monomials of degree d and by $Mon_{\leq d}(\underline{X})$ the set of monomials of degree at most d :

$$Mon_d(\underline{X}) := \{x_1^{e_1} \cdots x_n^{e_n} \mid e_i \geq 0, e_1 + \cdots + e_n = d\},$$

$$Mon_{\leq d}(\underline{X}) := \cup_{0 \leq i \leq d} Mon_i(\underline{X}).$$

Then the set A_d of all homogeneous polynomials of degree d is the vector space spanned by $Mon_d(\underline{X})$ over $\mathbb{F}$, and the set $A_{\leq d}$ of all polynomials of degree at most d is the vector space spanned by $Mon_{\leq d}(\underline{X})$ over $\mathbb{F}$:

$$A_d := \langle Mon_d(\underline{X}) \rangle_{\mathbb{F}}, \quad A_{\leq d} := \langle Mon_{\leq d}(\underline{X}) \rangle_{\mathbb{F}}.$$

Additional notations related to the multi-degree are introduced in Section 4.1.

2.2 Gröbner basis

Definition 1 ([3,8]). *We define the followings:*

1. A well-order $\succ$ on the monomials $Mon(\underline{X})$ is called a monomial order if it satisfies the following conditions:
 - i) $\forall u \in Mon(\underline{X}), u \succ 1$,
 - ii) $u_1 \succ u_2 \Rightarrow u_1 u_3 \succ u_2 u_3 (\forall u_3 \in Mon(\underline{X}))$.
2. Fix a monomial order $\succ$. For a polynomial $f = \sum_{(i_1, \dots, i_n)} a_{(i_1, \dots, i_n)} x_1^{i_1} \cdots x_n^{i_n} \in A$, we define the leading monomial $LM_{\succ}(f)$ as $LM_{\succ}(f) = \max_{\succ} \{x_1^{i_1} \cdots x_n^{i_n} \mid a_{(i_1, \dots, i_n)} \neq 0\}$. Moreover, for a set $S \subseteq A$, we define $LM_{\succ}(S)$ as $LM_{\succ}(S) = \{LM_{\succ}(f) \mid f \in S\}$. Here, we simply write $LM(f)$ and $LM(S)$ if $\succ$ is clear from the context.
3. Let I be an ideal of A . For a monomial order $\succ$, the finite set $G \subseteq I$ is called a Gröbner basis with respect to $\succ$ if for any $f \in I$, there exists $g \in G$ such that $LM_{\succ}(g) \mid LM_{\succ}(f)$.

Definition 2 ([3], p.272). *For an ideal I , the complement set of $LM_{\succ}(I)$ in $Mon(\underline{X})$ is called the set of reduced monomials and we denote it by $RM_{\succ}(I)$, or simply $RM(I)$ if $\succ$ is clear from the context, i.e.*

$$Mon(\underline{X}) = RM_{\succ}(I) \sqcup LM_{\succ}(I). \quad (1)$$

By the following remark, we have $\#RM_{\succ}(I) < \infty$ if $\dim_{\mathbb{F}} A/I < \infty$.

Remark 1 ([3], Proposition 6.52). An image of $RM_{\succ}(I)$ is a basis of the vector space A/I and there exists the following isomorphism of vector spaces:

$$\langle RM_{\succ}(I) \rangle_{\mathbb{F}} \cong A/I.$$

The following result is well known, but we include a proof to make the paper self-contained.

Lemma 1. *Let I be an ideal and $\succ$ be a monomial order. For a monomial $x_1^{e_1} \cdots x_n^{e_n} \in RM_\succ(I)$, we have $x_1^{e_1} \cdots x_j^{e_j-1} \cdots x_n^{e_n} \in RM_\succ(I)$ if $e_j \geq 1$.*

Proof. Suppose that there is $1 \leq j \leq n$ such that $e_j \geq 1$ and $u := x_1^{e_1} \cdots x_j^{e_j-1} \cdots x_n^{e_n} \notin RM_\succ(I)$. By Equation (1), $u \in LM_\succ(I)$ and there exists $f \in I$ such that $LM(f) = u$. Since $x_j f \in I$, we have the contradiction $x_1^{e_1} \cdots x_n^{e_n} = LM_\succ(x_j f) \in LM_\succ(I)$. $\square$

Corollary 1. *Let I be an ideal and $\succ$ be a monomial order. If $RM_\succ(I) \neq \emptyset$, then $1 \in RM_\succ(I)$.*

Proof. Immediate from Lemma 1. $\square$

Corollary 2. *Let I be an ideal such that $RM_\succ(I) \neq \emptyset$ and $\dim_{\mathbb{F}} A/I < \infty$. Then $\max\{\deg u \mid u \in RM_\succ(I)\} \leq \#RM_\succ(I) - 1$.*

Proof. By Corollary 1, we have $1 = x_1^0 \cdots x_n^0 \in RM_\succ(I)$. By Lemma 1, a monomial $x_1^{e_1} \cdots x_n^{e_n}$ can be obtained by multiplying 1 by at most $\#RM_\succ(I) - 1$ variables. Hence $\sum_i e_i \leq \#RM_\succ(I) - 1$. $\square$

2.3 Macaulay matrix

Let F be a finite set in $A = \mathbb{F}[x_1, \dots, x_n]$. In this paper, we consider the subspace $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ of the ideal $\langle F \rangle_A$:

Definition 3. *We define $[A \cdot F]_{\leq d} = \{g \cdot f \mid f \in F, g \in A_{\leq d - \deg f}\}$ and*

$$\begin{aligned} \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} &:= \langle g \cdot f \mid f \in F, g \in A_{\leq d - \deg f} \rangle_{\mathbb{F}} \\ &= \langle g \cdot f \mid f \in F, g \in Mon_{\leq d - \deg f}(\underline{X}) \rangle_{\mathbb{F}}. \end{aligned}$$

Remark 2. Since $\#F, \#Mon_{\leq d - \deg f}(\underline{X}) < \infty$, the cardinality of $\{u \cdot f \mid f \in F, u \in Mon_{\leq d - \deg f}(\underline{X})\}$ is finite and the dimension of the vector space $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ is finite.

In the remainder of this subsection, we introduce a method for computing a certain basis of the vector space $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$.

Definition 4. *Let U be a finite subset of $Mon(\underline{X})$ and let $\succ$ be a total order on U . Let $u_1, \dots, u_N$ be the elements of U ordered so that $u_i \succ u_{i+1}$ for all i .*

1. For $f \in A$, we define

$$\text{vect}_{\tilde{U}}(f) := (\text{Coeff}(f, u_1), \dots, \text{Coeff}(f, u_N)) \in \mathbb{F}^N,$$

where $\text{Coeff}(f, u)$ is the coefficient of u in f .

2. For a finite set $F \subset A$, fix an arbitrary enumeration $F = (f_1, \dots, f_m)$. We define $\text{Mac}(F, U; \succ)$ as the matrix whose i -th row is $\text{vect}_U^\succ(f_i)$, i.e.

$$\text{Mac}(F, U; \succ) := \begin{pmatrix} \text{Coeff}(f_1, u_1) & \cdots & \text{Coeff}(f_1, u_N) \\ \vdots & \ddots & \vdots \\ \text{Coeff}(f_m, u_1) & \cdots & \text{Coeff}(f_m, u_N) \end{pmatrix} \in \text{Mat}_{m \times N}(\mathbb{F}).$$

A different enumeration of F only permutes the rows of $\text{Mac}(F, U; \succ)$.

Let U be a finite set of monomials such that it contains all monomials in F , i.e.

$$U \supseteq \{u \in \text{Mon}(\underline{X}) \mid \exists f \in F \text{ s.t. } \text{Coeff}(f, u) \neq 0\}.$$

Suppose a total order $\succ$ is a monomial order. Since the monomials corresponding to the columns of $\text{Mac}(F, U; \succ)$ are ordered by the monomial order $\succ$, the pivots in an echelon form of $\text{Mac}(F, U; \succ)$ correspond to the leading monomials $LM_\succ(\langle F \rangle_{\mathbb{F}})$:

$$\begin{pmatrix} 1 & * & \cdots & & * \\ & 1 & * & \cdots & * \\ & & \ddots & & \vdots \\ & & & 1 & * \cdots * \end{pmatrix}.$$

Here, the pivots in the above figure are normalized. The following lemma was formulated at least as early as [12].

Lemma 2 (cf. Theorem 2.1, [12]). Fix a monomial order $\succ$. Let $F \subseteq A$ be a finite set. Then there exists a basis $\{g_1, \dots, g_r\}$ of $\langle F \rangle_{\mathbb{F}}$ such that

$$\{LM_\succ(g_i) \mid 1 \leq i \leq r\} = LM_\succ(\langle F \rangle_{\mathbb{F}}). \quad (2)$$

In particular, we have $\dim \langle F \rangle_{\mathbb{F}} = \sharp LM_\succ(\langle F \rangle_{\mathbb{F}})$.

Definition 5. Let $F \subseteq A$ be a finite set and $\succ$ be a monomial order. We call a basis $\{g_1, \dots, g_r\} \subseteq \langle F \rangle_{\mathbb{F}}$ satisfying (2) in Lemma 2 an echelon form (EF) basis of $\langle F \rangle_{\mathbb{F}}$ with respect to $\succ$.

Definition 6. Let F be a finite subset of A and $\succ$ be a monomial order. For a positive integer d , fix an arbitrary enumeration of the set $\{u \cdot f \mid f \in F, u \in \text{Mon}_{\leq d - \deg f}(\underline{X})\}$. The Macaulay matrix of this set with respect to $\succ$ on $\text{Mon}_{\leq d}(\underline{X})$ is called the degree d Macaulay matrix of F , and we denote it by $\text{Mac}_d^\succ(F)$, or simply by $\text{Mac}_d(F)$ if $\succ$ is clear from the context:

$$\text{Mac}_d^\succ(F) := \text{Mac}(\{u \cdot f \mid f \in F, u \in \text{Mon}_{\leq d - \deg f}(\underline{X})\}, \text{Mon}_{\leq d}(\underline{X}); \succ).$$

For a given value d , Algorithm 1 outputs an EF basis of $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ by computing an echelon form of $\text{Mac}_d(F)$:

Algorithm 1 (cf.[7]) Macaulay Matrix Generation and Echelon Form Computation

Input: $d \geq 0$, $\succ$: a monomial order, F : a finite subset in $A = \mathbb{F}[x_1, \dots, x_n]$

Output: The polynomials which correspond to the non-trivial rows in an echelon form of $Mac_d(F)$, i.e. an EF basis of $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$

- 1: For each $f \in F$ and for each monomial u with $\deg u \leq d - \deg f$, compute $u \cdot f$. Construct the Macaulay matrix $Mac_d(F)$ from the set $\{u \cdot f \mid f \in F, u \in Mon_{\leq d - \deg f}(\underline{X})\}$.
 - 2: Compute an echelon form of $Mac_d(F)$ by the Gaussian elimination.
 - 3: Return polynomials which correspond to non-trivial rows of this echelon form.
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Remark 3. We can restate Lemma 2 as follows. Fix a non-negative integer d . For a finite subset F of $A = \mathbb{F}[x_1, \dots, x_n]$, the output G of Algorithm 1, which is an EF basis of $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$, is finite and satisfies $LM_{\succ}(G) = LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$.

Hence, we have the following lemma for Algorithm 1:

Lemma 3. *Let F be a polynomial system and fix a monomial order. Then, there exists d such that $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ contains a Gröbner basis of the ideal $\langle F \rangle_A$. In particular, Algorithm 1 with a sufficiently large d computes the Gröbner basis.*

Proof. Let $G = \{g_1, \dots, g_s\}$ be a Gröbner basis of the ideal $\langle F \rangle_A$. Since $g_i \in \langle F \rangle_A$, there are $h_1^{(i)}, \dots, h_m^{(i)}$ such that $g_i = h_1^{(i)}f_1 + \dots + h_m^{(i)}f_m$ for each $1 \leq i \leq s$. Hence, we have $g_1, \dots, g_s \in \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ where $d \geq \max\{\deg(h_j^{(i)}f_j) \mid 1 \leq i \leq s, 1 \leq j \leq m\}$. An EF basis of $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ contains polynomials whose leading monomials coincide with $LM(g_1), \dots, LM(g_s)$, and this EF basis forms a Gröbner basis. $\square$

Remark 4 (Courtois et al.'s XL). We fix an elimination monomial order. In [7], Courtois et al. assume that there exists a value D such that the output of Algorithm 1 with the input D contains a univariate polynomial and describe the XL method in [7] using this value D . For any input polynomial system, [9] mentions that it is reasonable to consider that their XL works as follows (see Remark 1 in [9]). It executes Algorithm 1 (using $Mac_d^{pract}(F)$ in Remark 5) with $d = 0$ and obtains a univariate polynomial, if any. Otherwise, it increases d and executes the algorithm again. As a result, it deduces a system with one fewer variable by evaluating the input polynomial system at a solution to this univariate polynomial and obtains a solution to the input system by repeating this process. In this paper, we refer to the process as the *incremental search based on Algorithm 1*, which requires outputting a univariate polynomial. However, [9] also mentions that the process does not always terminate (see Remark 4 in [9]).

Remark 5 (Practical Macaulay matrix). In Algorithm 1, for the set $XL_d(F) := \{u \cdot f \mid f \in F, u \in Mon_{\leq d - \deg f}(\underline{X})\}$, we consider Macaulay matrix $Mac_d(X) = Mac(XL_d(F), Mon_{\leq d}(\underline{X}); \succ)$. However, it is more practical to consider the coefficient matrix for terms with only non-zero coefficients, and we can consider an algorithm which computes an echelon form of $Mac_d^{pract}(F) := Mac(XL_d(F), U; \succ)$

where

$$U := \{u \in \text{Mon}(\underline{X}) \mid \exists f \in XL_d(F) \text{ s.t. } \text{Coeff}(f, u) \neq 0\}.$$

The output of this algorithm is the same as Algorithm 1.

3 Solving polynomial systems via a Macaulay matrix

In this subsection, we suppose that $c_{RM} := \sharp RM_{\succ}(\langle F \rangle) = \dim_{\mathbb{F}} A / \langle F \rangle_A < \infty$. Moreover, we define $LM_{\succ}(\langle F \rangle_A)_{\leq d}$ as $LM_{\succ}(\langle F \rangle_A)_{\leq d} = LM_{\succ}(\langle F \rangle_A) \cap \text{Mon}_{\leq d}(\underline{X})$. Note that the case $c_{RM} = 0$ implies that the system of polynomial equations has no solution.

3.1 Gröbner basis computation

As mentioned in Section 3.2, to solve a system of polynomial equations, it is important to compute a Gröbner basis of the ideal generated by the defining polynomials. By Lemma 3, for a given monomial order, Algorithm 1 with a sufficiently large d outputs a Gröbner basis with respect to it. In this section, we investigate the values of d for which Algorithm 1 outputs a Gröbner basis.

We first present the simplest characterization, which applies to any monomial order:

Proposition 1. *Fix a monomial order. If there exists an element D in the set $\{d \geq 0 \mid \dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq c_{RM} \leq d\}$, then an EF basis of $\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}$ is a Gröbner basis. In particular, Algorithm 1 with D returns a Gröbner basis of $\langle F \rangle$.*

Proof. Since $c_{RM} \leq d$, it follows from Corollary 2 that $RM_{\succ}(\langle F \rangle) \subseteq \text{Mon}_{\leq d-1}(\underline{X})$. Thus $\text{Mon}_{\leq d}(\underline{X}) = RM_{\succ}(\langle F \rangle) \sqcup LM_{\succ}(\langle F \rangle)_{\leq d}$. It is obvious that $LM_{\succ}(\langle F \rangle)_{\leq d} \supseteq LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$, and we have $c_{RM} = \sharp RM_{\succ}(\langle F \rangle) = \sharp \text{Mon}_{\leq d}(\underline{X}) - \sharp LM_{\succ}(\langle F \rangle)_{\leq d} \leq \sharp \text{Mon}_{\leq d}(\underline{X}) - \sharp LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$ by Lemma 2. Since $\dim_{\mathbb{F}} \mathbb{F}[x_1, \dots, x_n]_{\leq d} - \dim_{\mathbb{F}} \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq c_{RM}$, i.e. $\sharp \text{Mon}_{\leq d}(\underline{X}) - \sharp LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}) \leq c_{RM}$, we have $LM_{\succ}(\langle F \rangle)_{\leq d} = LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$. Moreover, since $RM_{\succ}(\langle F \rangle) \cap \text{Mon}_d(\underline{X}) = \emptyset$, we obtain that $\text{Mon}_d(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$. Thus, every monomial of degree $> d$ is divided by some element of $LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$. Let G be an EF basis of $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$, and we have $LM_{\succ}(G) = LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$ by Remark 3. Since $LM_{\succ}(\langle F \rangle) = LM_{\succ}(\langle F \rangle)_{\leq d} \sqcup LM_{\succ}(\langle F \rangle)_{> d}$, for any $u \in LM_{\succ}(\langle F \rangle)$, there exists $g \in G$ such that $LM_{\succ}(g) \mid u$. Namely, G is a Gröbner basis. $\square$

Remark 6. By the proof of Proposition 1, if there is d such that $d \geq c_{RM}$ and $\dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq c_{RM}$, then it holds that $\dim_{\mathbb{F}} A_{\leq d'} / \langle [A \cdot F]_{\leq d'} \rangle_{\mathbb{F}} = c_{RM}$ for any $d' \geq d$, and vice versa.

In general, the set in Proposition 1 may be empty. In the following proposition, we show that the set is non-empty if there exists d such that $\text{Mon}_d(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$ (see Remark 6).

Proposition 2. Suppose $c_{RM} := \dim_{\mathbb{F}} A / \langle F \rangle_A < \infty$. The following conditions are equivalent:

1. There exists an integer d such that $Mon_d(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$.
2. There also exists d' such that $\dim_{\mathbb{F}} A_{\leq d'_0} / \langle [A \cdot F]_{\leq d'_0} \rangle_{\mathbb{F}} = c_{RM}$ ($\forall d'_0 \geq d'$).

Proof. $1 \Rightarrow 2$) Let d be an integer such that $Mon_d(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$. Hence, we have

$$Mon_{d'}(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d'} \rangle_{\mathbb{F}}) \quad (\forall d' \geq d). \quad (3)$$

This yields $RM_{\succ}(\langle F \rangle_A) \subseteq Mon_{\leq d-1}(\underline{X})$ by Equation (1), and it holds that $Mon_{\leq d-1}(\underline{X}) = LM_{\succ}(\langle F \rangle_A)_{\leq d-1} \sqcup RM_{\succ}(\langle F \rangle_A)$. For each element u of $LM_{\succ}(\langle F \rangle_A)_{\leq d-1}$, we fix one sequence $(h_1^{(u)}, \dots, h_m^{(u)})$ such that $LM_{\succ}(h_1^{(u)}f_1 + \dots + h_m^{(u)}f_m) = u$. Then we can choose $D \geq \max\{\deg h_i^{(u)}f_i \mid u \in LM_{\succ}(\langle F \rangle_A)_{\leq d-1}\}$ since $\sharp LM_{\succ}(\langle F \rangle_A)_{\leq d-1} < \infty$. Then, it holds that $LM_{\succ}(\langle F \rangle_A)_{\leq d-1} \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}})$. Moreover, by Equation (3), we have

$$\bigcup_{j=d}^D LM_{\succ}(\langle F \rangle_A)_j \subseteq \bigcup_{j=d}^D Mon_j(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}).$$

Thus, we obtain $LM_{\succ}(\langle F \rangle_A)_{\leq D} \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}})$ and it follows that $\dim A_{\leq D} / \langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}} = \sharp RM_{\succ}(\langle F \rangle_A) = c_{RM}$.

$2 \Rightarrow 1$) Since $c_{RM} < \infty$, we can take a sufficiently large d such that $c_{RM} \leq d$ and $d' \leq d$. Then, by the proof of Proposition 1, we have $Mon_d(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$. $\square$

In particular, we have the following proposition:

Proposition 3. Fix a monomial order. If there exists an element D in the set $\{d \geq 0 \mid \dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq c_{RM}, Mon_d(\underline{X}) \subseteq LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})\}$, then an EF basis of $\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}$ is a Gröbner basis. In particular, Algorithm 1 with D returns a Gröbner basis of $\langle F \rangle_A$.

Proof. Let d be an integer such that $Mon_d(\underline{X}) \subseteq LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$. Hence for $d' \geq d$, $Mon_{d'}(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d'} \rangle_{\mathbb{F}}) \subseteq LM(\langle F \rangle_A)$. In particular, we have $RM_{\succ}(\langle F \rangle_A) \subseteq Mon_{\leq d-1}(\underline{X})$ and $Mon_{\leq d}(\underline{X}) = LM(\langle F \rangle_A)_{\leq d} \sqcup RM_{\succ}(\langle F \rangle_A)$. Since $\dim_{\mathbb{F}} A_{\leq d} - \dim_{\mathbb{F}} \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq c_{RM}$, we obtain $c_{RM} = \sharp RM_{\succ}(\langle F \rangle_A) = \sharp Mon_{\leq d}(\underline{X}) - \sharp LM(\langle F \rangle_A)_{\leq d} \leq \sharp Mon_{\leq d}(\underline{X}) - \sharp LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}) \leq c_{RM}$ and $LM(\langle F \rangle_A)_{\leq d} = LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$. Hence, by Remark 3, an EF basis of $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ is a Gröbner basis. $\square$

Next, we will consider the graded version of Proposition 3. For a polynomial $f \in A$, we denote by f^{top} the highest homogeneous component of f . For a set F of polynomials, we define F^{top} as $F^{top} = \{f^{top} \mid f \in F\}$. Furthermore, for any $d \geq 0$, we define $\langle F^{top} \rangle_d = \langle h_1 f_1^{top} + \dots + h_m f_m^{top} \mid h_i \in A_{d - \deg f_i} \rangle_{\mathbb{F}}$.

Lemma 4. Fix a graded monomial order $\succ$. For $d \geq 0$, we have $A_d = \langle F^{top} \rangle_d$ if and only if $Mon_d(\underline{X}) \subseteq LM_{\succ}(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$.

Proof. The proof is straightforward and is omitted. $\square$

The following corollary is a version of Proposition 3 for degree-compatible monomial orders. It can be regarded as an analogue of a statement used in the proof of [16] for the graded reverse lexicographic order (see Remark 8).

Corollary 3. *Fix a graded monomial order. If there exists an element D in the set $\{d \geq 0 \mid \dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq c_{RM}, A_d = \langle F^{top} \rangle_d\}$, then an EF basis of $\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}$ is a Gröbner basis. In particular, Algorithm 1 with D returns a Gröbner basis of $\langle F \rangle_A$.*

Proof. This follows immediately from Proposition 3 and Lemma 4. $\square$

The following corollary may appear obvious from a geometric point of view (see Remark 7).

Corollary 4. *Suppose $c_{RM} := \dim_{\mathbb{F}} A / \langle F \rangle_A < \infty$. The following conditions are equivalent:*

1. *There exists d such that $\dim_{\mathbb{F}} A_{\leq d'} / \langle [A \cdot F]_{\leq d'} \rangle_{\mathbb{F}} = c_{RM}$ ($\forall d' \geq d$).*
2. *$\dim_{\mathbb{F}} A / \langle F^{top} \rangle_A < \infty$, i.e. there exists d such that $A_d = \langle F^{top} \rangle_d$.*

Proof. This follows immediately from Proposition 2 and Lemma 4.

When $\mathbb{F}$ is a finite field of order q , by adding the *field equations* $\{x_1^q - x_1, \dots, x_n^q - x_n\}$ to the polynomial system F , we have $\dim_{\mathbb{F}} A / \langle F \rangle_A < \infty$ and $\dim_{\mathbb{F}} A / \langle F^{top} \rangle_A < \infty$. In particular, there exists d such that $\dim_{\mathbb{F}} A_{\leq d'} / \langle [A \cdot F]_{\leq d'} \rangle_{\mathbb{F}} = c_{RM}$ for $d' \geq d$ by Corollary 4.

Remark 7. Corollary 4 can also be obtained as follows. In the projective variety V defined by homogenized polynomials of $f_1, \dots, f_m$, the condition $\dim_{\mathbb{F}} A / \langle F^{top} \rangle_A < \infty$ means that the points at infinity is empty. Since the Hilbert polynomial of the zero-dimensional variety V is the number of points counted with multiplicity, we have Corollary 4 by discussion in Appendix (see Appendix A).

Remark 8. Let x_0 be a new variable. We define $S = A[x_0]$ and obtain $F^h := \{f_1^h, \dots, f_m^h\} \subseteq S$ where f_i^h is the homogenization of f_i by x_0 . Then $\langle F^h \rangle_S$ is a homogeneous ideal and has the homogeneous component $\langle F^h \rangle_d$ of degree d , i.e. $\langle F^h \rangle_S = \bigoplus_d \langle F^h \rangle_d$. It follows that $\dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} = \dim_{\mathbb{F}} S_d / \langle F^h \rangle_d$ [9, 23] (see Appendix A).

1. We assume that F is a quadratic system of the type commonly considered in multivariate cryptography, satisfying $\dim_{\mathbb{F}} A / \langle F \rangle_A = 1$ and $\dim_{\mathbb{F}} A / \langle F^{top} \rangle_A < \infty$. By Remark 6, the smallest value of the set in Proposition 1 coincides with the *index of regularity* of $S / \langle F^h \rangle_S$, also referred to as the *Hilbert regularity* [10]. In particular, Proposition 1 shows that a Gröbner basis of $\langle F \rangle_A$ with respect to any monomial order can be computed by Algorithm 1 with the index of regularity of $S / \langle F^h \rangle_S$. By Serre's theorem (see,

e.g., Theorem 4.4.3 in [26]), the index of regularity of $S/\langle F^h \rangle_S$ is smaller than or equal to the *Castelnuovo-Mumford regularity* of $\langle F^h \rangle_S$, $\text{reg}(\langle F^h \rangle_S)$. In [6], they show that a Gröbner basis of $\langle F \rangle_A$ with respect to the graded reverse lexicographic (grevlex) order can be computed by Algorithm 1 with the regularity $\text{reg}(\langle F^h \rangle_S)$.

2. The set in Corollary 3 can be rewritten as $U := \{d \geq 0 \mid \dim_{\mathbb{F}} S_d / \langle F^h \rangle_d = c_{RM}, A_d = \langle F^{top} \rangle_d\}$. Corollary 3 shows that a Gröbner basis of $\langle F \rangle_A$ with respect to any graded monomial order can be computed by Algorithm 1 with any element in U . In [16], it is shown that the maximal degree of the elements of the reduced Gröbner basis of $\langle F^h \rangle_S$ with respect to the grevlex order is bounded above by the smallest value in U (see the proof of Theorem 30 in [16]). Using this fact and following an argument similar to [6], [18] states that a Gröbner basis of $\langle F \rangle_A$ with respect to the grevlex order can be computed by Algorithm 1 with the smallest value in U .

As will be seen later in the discussion of input degrees for the lexicographic order, under the conditions above, these bounds do not lead to smaller admissible degrees than those obtained in the classical analyses [23,9].

3.2 Motivation for computing a Gröbner basis

On solving a system of polynomial equations, the following proposition motivates the computation of a Gröbner basis of the ideal generated by the system.

Proposition 4 ([3], Theorem 6.54). *If $\dim_{\mathbb{F}} A / \langle F \rangle_A < \infty$, a Gröbner basis of $\langle F \rangle_A$ with respect to the lexicographic monomial order contains a univariate polynomial.*

Solving a univariate polynomial in Proposition 4 by Berlekamp algorithm [4] or its improvement, we can obtain a system with one less variable. Therefore, by applying Proposition 4 repeatedly, we can determine the solution to the system of polynomials.

In the Buchberger algorithm and its variants, a Gröbner basis computation with respect to the graded reverse lexicographic order is often efficient. Hence, we first construct a Gröbner basis with respect to the graded reverse lexicographic order and then convert it to that with respect to the lexicographic order using a monomial order conversion algorithm (such as FGLM [13] or improvements).

To obtain a system with one fewer variable, there is another approach, which finds a linear polynomial from the ideal $\langle F \rangle_A$. Indeed, if we can find a linear polynomial $a_n x_n + \dots + a_1 x_1 + a_0$, we obtain a system with one fewer variable by substituting x_i with $-(1/a_i) \cdot (a_n x_n + \dots + a_{i+1} x_{i+1} + a_{i-1} x_{i-1} + \dots + a_1 x_1 + a_0)$ where $a_i \neq 0$.

Based on the facts stated above, we can consider the following propositions for Algorithm 1 in order to obtain a system with one fewer variable:

Proposition 5 ([9,23,28]). *For the lexicographic monomial order, if there exists an element D in the set $\{d \geq 0 \mid \dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq d\}$, then an EF*

basis of $\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}$ contains a univariate polynomial. In particular, the output of Algorithm 1 with D contains a univariate polynomial.

Proof. We assume that $x_n \succ x_{n-1} \succ \dots \succ x_1$. The set $\text{Mon}_{\leq d}(\underline{X})$ contains $d+1$ elements $x_1^d, x_1^{d-1}, \dots, x_1, 1$. Since $\#\text{Mon}_{\leq d}(\underline{X}) - \#LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}) \leq d$, we have $LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}) \cap \{x_1^d, x_1^{d-1}, \dots, x_1, 1\} \neq \emptyset$. Hence, there exists $g \in \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ such that $LM(g) = x_1^e$ and g is a univariate polynomial in x_1 by definition of the lexicographic monomial order. $\square$

Remark 9. Note that we can replace the lexicographic monomial order in Proposition 5 by any elimination order.

The following proposition is commonly accepted in the implementation community:

Proposition 6.¹ *For the graded monomial order, if there exists an element D in the set $\{d \geq 1 \mid \dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq n\}$, then an EF basis of $\langle [A \cdot F]_{\leq D} \rangle_{\mathbb{F}}$ contains a polynomial of degree at most one. In particular, the output of Algorithm 1 with D contains a polynomial of degree at most one.*

Proof. Since $d \geq 1$, the set $\text{Mon}_{\leq d}(\underline{X})$ contains $n+1$ elements $x_n, \dots, x_1, 1$. Since $\#\text{Mon}_{\leq d}(\underline{X}) - \#LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}) \leq n$, we have $LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}) \cap \{x_n, \dots, x_1, 1\} \neq \emptyset$. Hence, there exists $g \in \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ such that $LM(g) = x_i$ or 1, and g is a polynomial of degree at most one by definition of the graded monomial order. $\square$

Remark 10. As mentioned in Remark 4, Diem [9] points out that the incremental search based on Algorithm 1, which requires outputting a univariate polynomial, does not necessarily terminate for every polynomial system. When the projective variety defined by homogenized polynomials of $f_1, \dots, f_m$ is zero-dimensional, Moh [23] shows that their XL terminates by proving the set in Proposition 5 is non-empty (see also [9, 28]). We will obtain the termination when $\dim_{\mathbb{F}} A / \langle F \rangle_A < \infty$; that is when the affine variety defined by $f_1, \dots, f_m$ is zero-dimensional. In fact, by Proposition 4, a Gröbner basis of $\langle F \rangle_A$ with respect to the lexicographic monomial order contains a univariate polynomial. By Lemma 3, Algorithm 1 with a sufficiently large d computes this Gröbner basis, and the incremental search based on Algorithm 1 terminates. Note that an affine variety remains zero-dimensional even for the resulting polynomial system after fixing a variable.

Remark 11. i) By the same discussion as [23], we see that the set of Proposition 6 is non-empty when the projective variety defined by homogenized polynomials of $f_1, \dots, f_m$ is zero-dimensional and the number of points $\tilde{c}$ counted with multiplicity is at most n . Thus, Algorithm 1 with a sufficiently large d outputs a linear polynomial.

¹ To the best of our knowledge, this fact is commonly accepted in the implementation community, although we have not found a direct citation.

- ii) When we consider the incremental search based on Algorithm 1, we do not always obtain a system with one less variable by finding only a linear polynomial. In fact, if the number of points $\tilde{c}$ in i) becomes larger than the number of variables after some reductions, comment i) above does not work. Then, we have to use a different approach, such as finding a univariate polynomial.
- iii) In the case $\tilde{c} = 1$, e.g. $c = 1$ and $\dim_{\mathbb{F}} A/\langle F^{top} \rangle < \infty$ (see Corollary 4), the incremental search based on Algorithm 1 can solve a polynomial system by finding linear polynomials.

3.3 Solving a system via kernel vector computation

In this section, we describe an algorithm already known from [29] in an explicit form and provide a reformulation.

We denote $\text{vect}_d^{\text{mon}}(\mathbf{a})$ by denote the row vector consisting of all monomials of degree at most d in n variables, evaluated at the point $\mathbf{a} \in \mathbb{F}^n$, and ordered according to a fixed monomial order. For the purpose of explanation, we fix the graded reverse lexicographic order as the monomial order. Namely, $\text{vect}_d^{\text{mon}}(\mathbf{a}) := (a_1^d, a_1 a_2^{d-1}, \dots, a_n^d, \dots, a_1 a_2 \dots a_n, 1)$ for $\mathbf{a} = (a_1, \dots, a_n)$ and $d \geq 1$. Furthermore, we assume $\deg f_1 = \dots = \deg f_m = 2$.

Let $\mathbf{a} = (a_1, \dots, a_n)$ be a root of a polynomial

$$f(x_1, \dots, x_n) = c_{11}x_1^2 + c_{12}x_1x_2 + \dots + c_{nn}x_n^2 + c_1x_1 + c_2x_2 + \dots + c_nx_n + c_0.$$

Then, for $\text{vect}_{\text{Mon}_{\leq 2}(\underline{X})}(f) = (c_{11}, c_{12}, \dots, c_{nn}, c_1, c_2, \dots, c_n, c_0)$ (see Definition 4), we have

$$\begin{aligned} \text{vect}_{\text{Mon}_{\leq 2}(\underline{X})}(f) \cdot {}^t\text{vect}_2^{\text{mon}}(\mathbf{a}) &= (c_{11}, c_{12}, \dots, c_{nn}, c_1, c_2, \dots, c_n, c_0) \begin{pmatrix} a_1^2 \\ a_1 a_2 \\ \vdots \\ a_n^2 \\ a_1 \\ a_2 \\ \vdots \\ a_n \\ 1 \end{pmatrix} \\ &= f(a_1, \dots, a_n) = 0. \end{aligned}$$

Similarly, for a common root $\mathbf{a} = (a_1, \dots, a_n)$ of a polynomial system $F = \{f_1, \dots, f_m\}$, we have

$$\text{Mac}(F, \text{Mon}_{\leq 2}(\underline{X})) \cdot {}^t\text{vect}_2^{\text{mon}}(\mathbf{a}) = \begin{pmatrix} f_1(a_1, \dots, a_n) \\ \vdots \\ f_m(a_1, \dots, a_n) \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Namely, ${}^t\text{vect}_2^{\text{mon}}(\mathbf{a})$ is a right kernel vector of $\text{Mac}(F, \text{Mon}_{\leq 2}(\underline{X}))$.

Moreover, for a monomial $u = x_1^{e_1} \cdots x_n^{e_n} \in \text{Mon}_{d-2}(\underline{X})$, the vector $\mathbf{a} = (a_1, \dots, a_n)$ is also a root of the polynomial uf_i of degree d :

$$(u \cdot f_i)(a_1, \dots, a_n) = a_1^{e_1} \cdots a_n^{e_n} \cdot f(a_1, \dots, a_n) = 0.$$

Thus, we have

$$\text{Mac}_d(F) \cdot {}^t\text{vect}_d^{\text{mon}}(\mathbf{a}) = \mathbf{0}.$$

In general, we obtain the following lemma:

Lemma 5. *Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{F}^n$ be a solution to $F = \mathbf{0}$ and $\succ$ be a total order. The right kernel of the Macaulay matrix $\text{Mac}_d(F)$ contains the non-trivial vector $\text{vect}_d^{\text{mon}}(\mathbf{a}) = (u_1(\mathbf{a}), \dots, u_N(\mathbf{a}))$ where $\text{Mon}_{\leq d}(\underline{X}) = \{u_1, \dots, u_N\}$ and $u_i \succ u_{i+1}$ ($1 \leq i \leq N-1$).*

Then, we can introduce Algorithm 2 for solving a system of polynomial equations over a finite field. This algorithm appears as early as [29], where the Wiedemann algorithm [27] is used in Step 2 of Algorithm 2 to search for the right kernel.

Algorithm 2 (cf.[29]) Macaulay Matrix Generation and Kernel Vector Computation

Input: $d \geq 1$, $q < \infty$, $\succ$: a total order, F : a finite subset in $A = \mathbb{F}_q[x_1, \dots, x_n]$ such that $F = \mathbf{0}$ is consistent over $\mathbb{F}_q$.

Output: solutions to $F = \mathbf{0}$

- 1: For each $f \in F$ and for each monomial u with $\deg u \leq d - \deg f$, compute $u \cdot f$. Construct the Macaulay matrix $\text{Mac}_d(F)$ from the set $\{u \cdot f \mid f \in F, u \in \text{Mon}_{\leq d - \deg f}(\underline{X})\}$.
 - 2: Compute (a basis of) the right kernel space for $\text{Mac}_d(F)$ by the Gaussian elimination.
 - 3: For each kernel vector $\mathbf{v}$ with $v_1 \neq 0$, return $(v_{x_1}/v_1, \dots, v_{x_n}/v_1)$ if it is a solution to $F = \mathbf{0}$ where v_u is a component of $\mathbf{v}$ which corresponds to the monomial $u \in \text{Mon}_{\leq d}(\underline{X})$.
-

Corollary 5. *Fix a total order. For any positive integer d , Algorithm 2 returns solutions to $F = \mathbf{0}$. In particular, Algorithm 2 terminates.*

Proof. This follows from the discussion before Lemma 5. □

The complexity of Algorithm 2 depends on the right kernel search on $\text{Mac}_d(F)$, i.e. Step 3 in Algorithm 2. When given the information $\{\dim_{\mathbb{F}} A_{\leq i} / \langle [A \cdot F]_{\leq i} \rangle_{\mathbb{F}}\}_i$, we can determine the dimension of the right kernel of $\text{Mac}_i(F)$ since

$$\begin{aligned} \dim(\text{Ker}^{\text{right}}(\text{Mac}_d(F))) &= \sharp \text{Mon}_{\leq d}(\underline{X}) - \text{Rank}(\text{Mac}_d(F)) \\ &= \dim_{\mathbb{F}} A_{\leq d} - \dim_{\mathbb{F}} \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}. \end{aligned} \tag{4}$$

We exploit this fact and give complexity estimation for certain polynomial systems in Section 5.

If $\dim \text{Ker}(\text{Mac}_d(F)) = 1$, its non-trivial vector is equal to $\text{vect}_d^{\text{mon}}(\mathbf{a})$ up to a constant multiple, and we do not need to search for a correct non-trivial vector in Step 3 of Algorithm 2. In this case, Algorithm 2 does not depend on the order of a finite field, and in particular, Algorithm 2 terminates for any field. Therefore, the following version is often considered.

Proposition 7. *Let $\mathbb{F}$ be a field. Assume $\dim_{\mathbb{F}} A/\langle F \rangle_A = 1$. If the set $\{d \geq 1 \mid \dim A_{\leq d}/\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} = 1\}$ is non-empty, Algorithm 2 with its element returns the solution of $F = 0$.*

Proof. This follows directly from the preceding discussion and the discussion before Lemma 5. $\square$

Moreover, by Equation (4), we can restate Lemma 5 as follows:

Lemma 6. *If $\dim_{\mathbb{F}} A/\langle F \rangle_A \geq 1$, then $\dim_{\mathbb{F}} A_{\leq d}/\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \geq 1$ for every d .*

Proof. This follows directly from Lemma 5 together with the preceding discussion. $\square$

Note that for the algebraic closure $\overline{\mathbb{F}}$ of $\mathbb{F}$, it holds that $\dim_{\mathbb{F}} A_{\leq d}/\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} = \dim_{\overline{\mathbb{F}}} \overline{A}_{\leq d}/\langle [\overline{A} \cdot F]_{\leq d} \rangle_{\overline{\mathbb{F}}}$ where $\overline{A} = \overline{\mathbb{F}} \otimes A$.

4 Solving polynomial systems via a Macaulay matrix with the multi-degree

As the previous section, we assume $c_{RM} := \sharp RM_{\succ}(\langle F \rangle) = \dim_{\mathbb{F}} A/\langle F \rangle_A < \infty$.

The consistent system of bilinear equations does not satisfy the condition $\dim_{\mathbb{F}} A/\langle F^{\text{top}} \rangle_A < \infty$, i.e. the sets in Proposition 1 and 3 are empty, but we can often obtain a Gröbner basis by using Algorithm 1 with some value d . Moreover, although the condition can be forcibly satisfied by adding the field equations $\{x_i^q - x_i\}_i$, Algorithm 1 often produces a Gröbner basis with a smaller value d than the minimum value of the set of Proposition 1 and 3. In this section, we introduce multi-degree algorithms to illustrate this fact and provide the foundations for the multi-degree case from the previous section.

4.1 Macaulay matrix with the multi-degree

For $\mathbf{d} = (d_1, \dots, d_s)$ and $\mathbf{d}' = (d'_1, \dots, d'_s)$, while it might be more appropriate to denote by $\mathbf{d} \leq^s \mathbf{d}'$ the condition $d_i \leq d'_i$ for each i , in this paper, we will simply write $\mathbf{d} \leq \mathbf{d}'$. Moreover, we write $\mathbf{d} \leq \mathbf{d}'$ when $\mathbf{d} \leq \mathbf{d}'$ and $\mathbf{d} \neq \mathbf{d}'$.

Let $\underline{X} = \underline{X}_1 \cup \dots \cup \underline{X}_s$ where $\underline{X}_i = \{x_{1,i}, \dots, x_{n_i,i}\}$. For a monomial $u = x_{1,1}^{e_{1,1}} \dots x_{n_1,1}^{e_{n_1,1}} \dots x_{1,s}^{e_{1,s}} \dots x_{n_s,s}^{e_{n_s,s}}$ in $A = \mathbb{F}[\underline{X}] = \mathbb{F}[\underline{X}_1, \dots, \underline{X}_s]$, we define degrees $\deg_{\underline{X}_i}$ and $\deg_{\mathbb{Z}_{\geq 0}^s}$ as follows:

$$\deg_{\underline{X}_i} u = e_{1,i} + \dots + e_{n_i,i} \in \mathbb{Z}_{\geq 0},$$

$$\deg_{\mathbb{Z}_{\geq 0}^s} u = (\deg_{\underline{X}_1} u, \dots, \deg_{\underline{X}_s} u) \in \mathbb{Z}_{\geq 0}^s.$$

Moreover, for a polynomial $f = \sum_{u \in \text{Mon}(\underline{X})} c_u \cdot u$ in A , we define $\deg_{\underline{X}_i} f = \max\{\deg_{\underline{X}_i} u \mid c_u \neq 0\}$ and $\deg_{\mathbb{Z}_{\geq 0}^s} f = (\deg_{\underline{X}_1} f, \dots, \deg_{\underline{X}_s} f) \in \mathbb{Z}_{\geq 0}^s$.

We define $\text{Mon}_{(d_1, \dots, d_s)}(\underline{X}) = \{u \in \text{Mon}(\underline{X}) \mid \deg_{\mathbb{Z}_{\geq 0}^s} u = (d_1, \dots, d_s)\}$, $\text{Mon}_{\leq (d_1, \dots, d_s)}(\underline{X}) = \bigcup_{i_j \leq d_j} \text{Mon}_{(i_1, \dots, i_s)}(\underline{X})$ where $\text{Mon}_{(d_1, \dots, d_s)}(\underline{X}) = \{0\}$ if $d_i < 0$. Moreover, we define $A_{\mathbf{d}} = \langle \text{Mon}(\underline{X})_{\mathbf{d}} \rangle_{\mathbb{F}}$, $A_{\leq \mathbf{d}} = \langle \text{Mon}_{\leq \mathbf{d}}(\underline{X}) \rangle_{\mathbb{F}}$.

In this section, we devote the following vector space:

$$\begin{aligned} \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} &:= \langle g \cdot f \mid f \in F, g \in A_{\leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f} \rangle_{\mathbb{F}} \\ &= \langle u \cdot f \mid f \in F, u \in \text{Mon}_{\leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f}(\underline{X}) \rangle_{\mathbb{F}}. \end{aligned}$$

Definition 7 below and Algorithm 3 (see [24]) are multi-degree analogues of $\text{Mac}_{\mathbf{d}}(F)$ and Algorithm 1, respectively, in order to compute an EF basis of $\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$ (see also Remark 12).

Definition 7. Let F be a finite subset of A and $\succ$ be a monomial order. For $\mathbf{d} \in \mathbb{Z}_{\geq 0}^s$, fix an arbitrary enumeration of the finite set $\{u \cdot f \mid f \in F, u \in \text{Mon}_{\leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f}(\underline{X})\}$. The Macaulay matrix of this set with respect to $\succ$ on $\text{Mon}_{\leq \mathbf{d}}(\underline{X})$ is called the degree $\mathbf{d}$ Macaulay matrix of F , and we denote it by $\text{Mac}_{\mathbf{d}}^{\succ}(F)$, or simply by $\text{Mac}_{\mathbf{d}}(F)$ if $\succ$ is clear from the context:

$$\text{Mac}_{\mathbf{d}}^{\succ}(F) := \text{Mac}(\{u \cdot f \mid f \in F, u \in \text{Mon}_{\leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f}(\underline{X})\}, \text{Mon}_{\leq \mathbf{d}}(\underline{X}); \succ).$$

Algorithm 3 Multi-Degree Macaulay Matrix Generation and Echelon Form Computation

Input: $\mathbf{d} \in \mathbb{Z}_{\geq 0}^s$, $\succ$: a monomial order, F : a finite subset in $A = \mathbb{F}[\underline{X}_1, \dots, \underline{X}_s]$

Output: The polynomials which correspond to the non-trivial rows in an echelon form of $\text{Mac}_{\mathbf{d}}(F)$, i.e. an EF basis of $\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$

- 1: For each $f \in F$ and for each monomial u with $\deg_{\mathbb{Z}_{\geq 0}^s} u \leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f$, compute $u \cdot f$. Construct the Macaulay matrix $\text{Mac}_{\mathbf{d}}(F)$ from the set $\{u \cdot f \mid f \in F, u \in \text{Mon}_{\leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f}(\underline{X})\}$.
 - 2: Compute a echelon form of $\text{Mac}_{\mathbf{d}}(F)$ by the Gaussian elimination.
 - 3: Return polynomials which correspond to non-trivial rows of this echelon form.
-

Remark 12. An argument analogous to Remark 3 applies to Algorithm 3 as follows. Fix a vector $\mathbf{d} \in \mathbb{Z}_{\geq 0}^s$ and for a finite subset F of $A = \mathbb{F}[x_1, \dots, x_n]$, the output G of Algorithm 3, which is an EF basis of $\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$, is finite and satisfies $LM_{\succ}(G) = LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}})$.

Remark 13. We consider the termination of an incremental search based on Algorithm 3 as shown in Remark 4. Let $d \geq 0$. Then for $\mathbf{d} = (d_1, \dots, d_s)$ such that $d_i \geq d$ for all i , we have $\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \subseteq \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$. Hence, it follows the termination properties in Lemma 3 and Remark 10 and 11. If one intends to design an increasing search based on Algorithm 3 applicable to arbitrary inputs, a naive approach would be to exhaustively test all multi-degrees in order of increasing total degree.

4.2 Gröbner basis computation and admissible multi-degrees

Similar to Corollary 2, the following assertion holds:

Corollary 6. *For an ideal I , we have $\max\{\deg_{\underline{X}_i} u \mid u \in RM_{\succ}(I)\} \leq \sharp RM_{\succ}(I) - 1$ if $RM_{\succ}(I) \neq \emptyset$.*

The following proposition gives a simple characterization of admissible input degrees for arbitrary monomial orders in the multi-degree setting:

Proposition 8. *Fix a monomial order. If there exists an element $\mathbf{D}$ in the set $\{\mathbf{d} = (d_1, \dots, d_s) \mid \dim A_{\leq \mathbf{d}} / \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} \leq c_{RM} \leq d_i, 1 \leq \forall i \leq s\}$, then an EF basis of $\langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}}$ is a Gröbner basis. In particular, Algorithm 3 with $\mathbf{D}$ returns a Gröbner basis of $\langle F \rangle_A$.*

Proof. By Corollary 6, we have $RM_{\succ}(\langle F \rangle) \subseteq \cup_{i_j < d_j} Mon_{(i_1, \dots, i_s)}(X)$. In particular, it follows that

$$Mon_{\leq \mathbf{d}}(\underline{X}) \setminus \cup_{i_j < d_j} Mon_{(i_1, \dots, i_s)}(\underline{X}) \subseteq LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}). \quad (5)$$

Namely, each monomial in $Mon(X) \setminus Mon_{\leq \mathbf{d}}(X)$ is divided by some monomial in $LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}})$. Moreover, since $LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \subseteq LM(\langle F \rangle_A)_{\leq \mathbf{d}}$ and $c_{RM} = \sharp Mon_{\leq \mathbf{d}}(X) - \sharp LM(\langle F \rangle_A)_{\leq \mathbf{d}} \leq \sharp Mon_{\leq \mathbf{d}}(X) - \sharp LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \leq c_{RM}$, it follows that $LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) = LM(\langle F \rangle_A)_{\leq \mathbf{d}}$. By Remark 12, an EF basis of $\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$ is a Gröbner basis. $\square$

Essentially, the following proposition holds:

Proposition 9. *Fix a monomial order. Let $(\ell_1, \dots, \ell_s) = \{(d_1, \dots, d_s) \mid d_i = \max\{\deg_{\underline{X}_i} u \mid u \in RM_{\succ}(\langle F \rangle_A)\}\}$. If there exists an element $\mathbf{D}$ in the set $\{\mathbf{d} = (d_1, \dots, d_s) \mid \dim A_{\leq \mathbf{d}} / \langle A \cdot F \rangle_{\leq \mathbf{d}} \leq c_{RM}, d_i \geq \ell_i + 1 \ (1 \leq \forall i \leq s)\}$, then an EF basis of $\langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}}$ is a Gröbner basis. In particular, Algorithm 3 with $\mathbf{D}$ returns a Gröbner basis of $\langle F \rangle_A$.*

From the condition $d_i \geq \ell_i + 1$ for any i , we obtain

$$RM_{\succ}(\langle F \rangle) \subseteq \bigcup_{i_j < d_j} Mon_{(i_1, \dots, i_s)}(X)$$

which proves the proposition.

In Proposition 1 and 3, in order to show that any monomial of degree at least d is divided by some element of $LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}})$, we proved that

$$Mon_d(\underline{X}) \subseteq LM(\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}).$$

Equation (5) in the proof of Proposition 8 plays the same role as this inclusion condition. Based on this, in Definition 8, we introduce the subset $\partial Mon_{\leq d}(\underline{X})$ as a multi-degree analogue of $Mon_d(\underline{X}) (= Mon_{\leq d}(\underline{X}) \setminus Mon_{\leq d-1}(\underline{X}))$:

Definition 8. For $\mathbf{d} = (d_1, \dots, d_s) \in \mathbb{Z}_{\geq 0}^s$, we define

$$\partial Mon_{\leq \mathbf{d}}(\underline{X}) := Mon_{\leq \mathbf{d}}(\underline{X}) \setminus \cup_{i_j < d_j} Mon_{(i_1, \dots, i_s)}(\underline{X}). \quad (6)$$

Indeed, we have the following proposition:

Proposition 10. Fix a monomial order. If there exists an element $\mathbf{D}$ in the set $\{\mathbf{d} = (d_1, \dots, d_s) \mid \dim A_{\leq \mathbf{d}} / \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} \leq c_{RM}, \partial Mon_{\leq \mathbf{d}}(\underline{X}) \subseteq LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}})\}$, then an EF basis of $\langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}}$ is a Gröbner basis. In particular, Algorithm 3 with $\mathbf{D}$ returns a Gröbner basis of $\langle F \rangle_A$.

Proof. When $\partial Mon_{\leq \mathbf{d}}(\underline{X}) \subseteq LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}})$, by definition, we have that any monomial of $Mon(\underline{X}) \setminus Mon_{\leq \mathbf{d}}(\underline{X})$ is divided by some element of $LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}})$. Moreover, we have $RM_{\succ}(\langle F \rangle_A) \subseteq Mon_{\leq \mathbf{d}}(\underline{X})$ and $Mon_{\leq \mathbf{d}}(\underline{X}) = LM(\langle F \rangle_A)_{\leq \mathbf{d}} \sqcup RM(\langle F \rangle_A)$ by (6). Since $\sharp LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \subseteq \sharp LM(\langle F \rangle_A)_{\leq \mathbf{d}}$, we have $c_{RM} = \sharp RM(\langle F \rangle_A) = \sharp Mon_{\leq \mathbf{d}}(\underline{X}) - \sharp LM(\langle F \rangle_A)_{\leq \mathbf{d}} \leq \sharp Mon_{\leq \mathbf{d}}(\underline{X}) - \sharp LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \leq c_{RM}$ and $LM(\langle F \rangle_A)_{\leq \mathbf{d}} = LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}})$. By Remark 12, an EF basis of $\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$ is a Gröbner basis. $\square$

Similar to Proposition 2, the following lemma holds:

Lemma 7. If there is $\mathbf{d}$ such that $\partial Mon_{\leq \mathbf{d}}(\underline{X}) \subseteq LM(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}})$, then there is also $\mathbf{d}'_0$ such that $\dim A_{\leq \mathbf{d}'} / \langle [A \cdot F]_{\leq \mathbf{d}'} \rangle_{\mathbb{F}} = c_{RM}$ for any $\mathbf{d}'$ with $\mathbf{d}'_0 \leq \mathbf{d}'$.

Proof. The proof is straightforward. $\square$

4.3 Other algorithms and admissible multi-degrees

In this section, we provide multi-degree versions of the results in Sections 3.2 and 3.3.

To obtain a system with one fewer variable, we show two approaches using Algorithm 1 and can apply these to Algorithm 3. Namely, the following assertion holds:

Proposition 11. 1) Fix the lexicographic monomial order where $x_{ij} \succ x_{11}$ for any $(i, j) \neq (1, 1)$. If there exists an element $\mathbf{D}$ in the set $\{\mathbf{d} = (d_1, \dots, d_s) \in \mathbb{Z}_{\geq 0}^s \mid \dim_{\mathbb{F}} A_{\leq \mathbf{d}} / \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} \leq d_1\}$, then an EF basis of $\langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}}$ contains a univariate polynomial in $x_{1,1}$. In particular, the output of Algorithm 3 with $\mathbf{D}$ contains a univariate polynomial in $x_{1,1}$.

- 2) Fix a graded monomial order, and let $n_i = \#X_i$. If there exists an element $\mathbf{D}$ in the set $\{\mathbf{d} = (d_1, \dots, d_s) \in \mathbb{Z}_{\geq 0}^s \setminus \{\mathbf{0}\} \mid \dim_{\mathbb{F}} A_{\leq \mathbf{d}} / \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} \leq \sum_{d_i \geq 1} n_i\}$, then an EF basis of $\langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}}$ contains a polynomial of degree at most one. In particular, the output of Algorithm 3 with $\mathbf{D}$ contains a polynomial of degree at most one.

Proof. 1) Since $\{x_{11}^{d_1}, x_{11}^{d_1-1}, \dots, x_{11}, 1\} \subseteq \text{Mon}_{\leq \mathbf{d}}(\underline{X})$ and $\# \text{Mon}_{\leq \mathbf{d}}(\underline{X}) - \# \text{LM}(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \leq d_1$, we have $\text{LM}(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \cap \{x_{11}^{d_1}, x_{11}^{d_1-1}, \dots, x_{11}, 1\} \neq \emptyset$. Hence, there exists $g \in \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$ such that $\text{LM}(g) = x_{11}^e$ and g is a univariate polynomial in x_{11} by definition of the lexicographic monomial order.

2) By reordering the indices, we may assume that $d_1, \dots, d_\ell \geq 1$ and $d_{\ell+1} = \dots = d_s = 0$ ($\ell \leq s$) without loss of generality. Then, the set $\text{Mon}_{\leq \mathbf{d}}(\underline{X})$ contains $1 + n_1 + \dots + n_s$ elements $1, x_{1,1}, \dots, x_{n_1,1}, \dots, x_{1,\ell}, \dots, x_{n_\ell,\ell}$. Since $\# \text{Mon}_{\leq \mathbf{d}}(\underline{X}) - \# \text{LM}(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \leq \sum_{i=1}^\ell n_i$, we have $\text{LM}(\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}) \cap (\{1\} \cup \{x_{i,j}\}_{1 \leq j \leq \ell, 1 \leq i \leq n_j}) \neq \emptyset$. Hence, there exists $g \in \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}$ such that $\text{LM}(g) = x_{i,j}$ or 1, and g is a polynomial of degree at most one by definition of the graded monomial order. $\square$

Remark 14. Note that we can replace the lexicographic monomial order in 1) of Proposition 11 by an elimination order.

Based on the following statements, we can provide Algorithm 4 (see [25, 24, 5]) as its multi-degree version Algorithm 2. This algorithm has already been introduced in [25] for the case $s = 2$.

Lemma 8. Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{F}^n$ be a solution to $F = \mathbf{0}$ and $\succ$ be a total order. The right kernel of the Macaulay matrix $\text{Mac}_{\mathbf{d}}(F)$ contains the non-trivial vector $\text{vect}_{\mathbf{d}}^{\text{mon}}(\mathbf{a}) = (u_1(\mathbf{a}), \dots, u_N(\mathbf{a}))$ where $\text{Mon}_{\leq \mathbf{d}}(\underline{X}) = \{u_1, \dots, u_N\}$ and $u_i \succ u_{i+1}$ ($1 \leq i \leq N-1$).

Algorithm 4 Multi-Degree Macaulay Matrix Generation and Kernel Vector Computation

Input: $\mathbf{d} \in \mathbb{Z}_{\geq 1}^s$, $q < \infty$, $\succ$: a monomial order, F : a finite subset in $A = \mathbb{F}_q[\underline{X}_1, \dots, \underline{X}_s]$ such that $F = \mathbf{0}$ is consistent over $\mathbb{F}_q$.

Output: solutions to $F = \mathbf{0}$

- 1: For each $f \in F$ and for each monomial u with $\deg_{\mathbb{Z}_{\geq 0}^s} u \leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f$, compute $u \cdot f$. Construct the Macaulay matrix $\text{Mac}_{\mathbf{d}}(F)$ from the set $\{u \cdot f \mid f \in F, u \in \text{Mon}_{\leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f}(\underline{X})\}$.
 - 2: Compute (a basis of) the right kernel space for $\text{Mac}_{\mathbf{d}}(F)$ by the Gaussian elimination.
 - 3: For each kernel vector $\mathbf{v}$ with $v_1 \neq 0$, return $(v_{x_1}/v_1, \dots, v_{x_n}/v_1)$ if it is a solution to $F = \mathbf{0}$ where v_u is a component of $\mathbf{v}$ which corresponds to the monomial $u \in \text{Mon}_{\leq \mathbf{d}}(\underline{X})$.
-

Corollary 7. *Fix a total order. For any $\mathbf{d} \in \mathbb{Z}_{\geq 1}^s$, Algorithm 4 returns solutions to $F = \mathbf{0}$. In particular, Algorithm 4 terminates.*

Proof. This follows by applying the argument used before Lemma 5 to Algorithm 4. $\square$

In particular, the following proposition is often used to avoid the iteration in Step 3 of Algorithm 4.

Proposition 12. *Let $\mathbb{F}$ be a field. Assume $\dim_{\mathbb{F}} A/\langle F \rangle = 1$. If $\{\mathbf{d} \in \mathbb{Z}_{\geq 1}^s \mid \dim A_{\leq \mathbf{d}}/\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} = 1\}$ is non-empty, Algorithm 4 with its element returns the solution to $F = \mathbf{0}$.*

Proof. This follows directly from Lemma 8. $\square$

5 Discussion for applications

In this section, we restate the results from the previous sections for a specific polynomial system and provide an upper bound on the complexity of algorithms in those sections. This polynomial system has a generating function of the sequence $\{\dim_{\mathbb{F}} A_{\leq i}/\langle [A \cdot F]_{\leq i} \rangle_{\mathbb{F}}\}_{i \geq 0}$ which is determined by the degrees and the number of variables.

For explanation, we consider $F = \{f_1, \dots, f_m\} \subseteq \mathbb{F}[x_1, \dots, x_n]$ with $\deg f_1 = \dots = \deg f_m = 2$. We introduce the following notation:

Definition 9. *Let $a = \sum_{i \geq 0} a_i t^i$ be a formal power series over $\mathbb{Q}$, and let $c \in \mathbb{Z}_{\geq 0}$. Then, we define*

$$\left[\sum_{i \geq 0} a_i t^i \right]_{(c)} := \sum_{i \geq 0} b_i t^i, \quad b_i = \begin{cases} c & \text{if } \exists i_0 \in \{c, \dots, i\} \text{ s.t. } a_{i_0} \leq c \\ a_i & \text{otherwise} \end{cases}.$$

The symbol $[\cdot]_{(0)}$, i.e. the case $c = 0$ in Definition 9, has already been introduced in [15] and has been extensively studied in [28]. Note that if the polynomial system F has finitely many solutions, then, as shown in Lemma 6, $\dim_{\mathbb{F}} A_{\leq d}/\langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$ is strictly greater than zero. Furthermore, under the condition $\dim_{\mathbb{F}} A/\langle F^{top} \rangle < \infty$, this value coincides with the number of solutions $c_{RM} > 0$ for sufficiently large degrees d , as stated in Corollary 4. Therefore, the notation $[\cdot]_{(c)}$ for $c > 0$ is meaningful in Corollary 8 below.

In this paper, we treat Gaussian elimination as a linear algebra operation on Macaulay matrices. By Remark 15 below, we focus on the minimum values appearing in the sets defined in several propositions related to solving polynomial systems, namely Proposition 1, 3, 5, 6, 7, 8, 11, and 12.

Using the notation in Definition 9, we restate the statements in Section 3 and 4 as follows:

Corollary 8. Let $F = \{f_1, \dots, f_m\} \subseteq \mathbb{F}[x_1, \dots, x_n]$ with $\deg f_i = 2$ and $c = \dim_{\mathbb{F}} A / \langle F \rangle_A$. Suppose

$$\sum_{i \geq 0} \dim_{\mathbb{F}} A_{\leq i} / \langle [A \cdot F]_{\leq i} \rangle_{\mathbb{F}} = \left[\frac{(1-t^2)^m}{(1-t)^{n+1}} \right]_{(c)}. \quad (7)$$

Then the following assertions hold:

1. Let $D = \min\{d \geq c \mid \text{Coeff}((1-t^2)^m(1-t)^{-(n+1)}, t^d) \leq c\} < \infty$. For a monomial order, Algorithm 1 (using Gaussian elimination) outputs a Gröbner basis of the ideal $\langle F \rangle_A$ with at most

$$m \binom{n+D}{D}^w \quad (8)$$

multiplications and divisions, where $w = 3$.

2. Let $D = \min\{d \geq 0 \mid \text{Coeff}([(1-t^2)^m(1-t)^{-(n+1)}]_{(c)}, t^d) \leq d\} < \infty$. For a lexicographic monomial order, Algorithm 1 (using Gaussian elimination) computes univariate polynomials in the ideal $\langle F \rangle_A$ with at most (8) multiplications and divisions.
3. Let $D = \min\{d \geq 1 \mid \text{Coeff}([(1-t^2)^m(1-t)^{-(n+1)}]_{(c)}, t^d) \leq n\} < \infty$. For a graded monomial order, Algorithm 1 (using Gaussian elimination) computes polynomials of degree at most one in the ideal $\langle F \rangle_A$ with at most (8) multiplications and divisions.
4. Let $D \in \mathbb{Z}_{\geq 1}$ and $c_D = \text{Coeff}([(1-t^2)^m(1-t)^{-(n+1)}]_{(c)}, t^D)$. Assume $\mathbb{F}$ is a finite field of order q . For a monomial order, Algorithm 2 (using Gaussian elimination) outputs solutions of the system $F = \mathbf{0}$ with at most

$$m \binom{n+D}{D}^w + q^{c_D} \cdot 2 \cdot c_D \cdot m \cdot \binom{n+D}{D} \cdot \binom{n+2}{2}$$

multiplications and divisions.

Note that the assumption (7) and $D < \infty$ in the assertion 1 of Corollary 8 imply that the equivalent conditions in Corollary 4 hold.

Moreover, we also mention an analogue of the previous section for the multi-degree. Here, we only introduce the case that a part of the generating function of $\{\dim_{\mathbb{F}} A_{\leq \mathbf{i}} / \langle [A \cdot F]_{\leq \mathbf{i}} \rangle_{\mathbb{F}}\}_{\mathbf{i} \in \mathbb{Z}_{\geq 0}^s}$ is determined by the degrees and the number of variables.

In [25], they suppose that some polynomial systems solved in the Rainbow-Band-Separation attack against Rainbow with a parameter set (v, o_1, o_2) satisfy the assumption of Corollary 9 below at $s = 2$, $\sharp \underline{X}_1 = n_x$, $\sharp \underline{X}_2 = n_y$, $\deg_{\mathbb{Z}_{\geq 0}^s} f_1 = \dots = \deg_{\mathbb{Z}_{\geq 0}^s} f_{m_x} = (0, 2)$, and $\deg_{\mathbb{Z}_{\geq 0}^s} f_{m_x+1} = \dots = \deg_{\mathbb{Z}_{\geq 0}^s} f_{m_x+m_{xy}} = (1, 1)$ where $n_x = v + o_1$, $n_y = o_2$, $m_x = o_1 + o_2$, $m_{xy} = v + o_1 + o_2 - 1$ (see [25]). Moreover, in [5], they suppose that some polynomial systems solved in their new attack against SNOVA with a parameter set (v, o, ℓ) satisfy the assumption

of Corollary 9 at $s = \ell$, $\sharp \underline{X}_i = o\ell - k/\ell + 1$, $F = \cup_{1 \leq i \leq j \leq \ell} F_{ij}$, $\deg_{\mathbb{Z}_{\geq 0}^\ell} f_k^{(ij)} = \mathbf{e}_i + \mathbf{e}_j$ ($f_k^{(ij)} \in F_{ij}$), $\sharp F_{ij} = 2o$ (if $i \neq j$) or o (if $i = j$) where k is the number of fixed variables in the hybrid approach such that $\ell \mid k$.

Corollary 9 (cf. [25,5]). *Let $\mathbb{F}$ be a finite field of order q and let $F = \{f_1, \dots, f_m\} \subseteq \mathbb{F}[\underline{X}_1, \dots, \underline{X}_s]$ with $\deg_{\mathbb{Z}_{\geq 0}^s} f_i = \mathbf{d}_i = (d_1^{(i)}, \dots, d_s^{(i)})$ and $c = \dim_{\mathbb{F}} A/\langle F \rangle_A$. Suppose that $\mathbf{D}' \in \mathbb{Z}_{\geq c}^s$ satisfies for any $\mathbf{D} = (D_1, \dots, D_s) \leq \mathbf{D}'$,*

$$c_{\mathbf{D}} := \dim_{\mathbb{F}} A_{\leq \mathbf{D}} / \langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}} = \begin{cases} \text{Coeff}_c(G(t_1, \dots, t_s), t_1^{D_1} \dots t_s^{D_s}) & \text{if } \mathbf{D} \neq \mathbf{D}' \\ 1 & \text{if } \mathbf{D} = \mathbf{D}' \end{cases}$$

where

$$G(t_1, \dots, t_s) := \frac{\prod_{j=1}^m (1 - t_1^{d_1^{(j)}} \dots t_s^{d_s^{(j)}})}{(1 - t_1)^{n_1+1} \dots (1 - t_s)^{n_s+1}}.$$

Then, Algorithm 3 with the multi-degree $\mathbf{D}' \in \mathbb{Z}_{\geq c}^s$ (using Gaussian elimination) outputs a Gröbner basis of the ideal $\langle F \rangle_A$ with at most

$$m(M_{\mathbf{D}'})^w$$

multiplications and divisions, and for $\mathbf{D} \in \mathbb{Z}_{\geq 1}^s$ such that $\mathbf{D} \leq \mathbf{D}'$, Algorithm 4 with the multi-degree $\mathbf{D}$ (using Gaussian elimination) computes solutions to $F = \mathbf{0}$ with at most

$$m(M_{\mathbf{D}})^w + 2 \cdot q^{c_{\mathbf{D}}} \cdot c_{\mathbf{D}} \cdot M_{\mathbf{D}} \cdot \sum_{1 \leq j \leq m} M_{\mathbf{d}_j}$$

multiplications and divisions where $w = 3$ and $M_{\mathbf{D}} = \binom{n_1+D_1}{D_1} \dots \binom{n_s+D_s}{D_s}$.

In the following corollary, we consider a certain bilinear system; in particular, it does not satisfy $\dim_{\mathbb{F}} A/\langle F^{top} \rangle_A < \infty$. For example, in [1], they suppose that some polynomial systems solved in the support minors method against the MinRank problem, which is a problem of finding a linear combination of k matrices of size $m \times n$ whose rank is at most r , satisfy the assumption of Corollary 10:

Corollary 10 (cf. [1]). *Let $\mathbb{F}$ be a finite field of order q and m, n, r , and k be positive integers. Let $F = \{f_1, \dots, f_{m \cdot \binom{n}{r+1}}\} \subseteq \mathbb{F}[\underline{X}_1, \underline{X}_2]$ where $\deg_{\mathbb{Z}_{\geq 0}^2} f_i = (1, 1)$, $\sharp \underline{X}_1 = m$, $\sharp \underline{X}_2 = \binom{n}{r}$, and $\dim_{\mathbb{F}} A/\langle F \rangle_A = 1$. We define*

$$C_{(b)} = \binom{n}{r} \binom{k+b-1}{b} - \sum_{i=1}^b (-1)^{i+1} \binom{n}{r+i} \binom{m+i-1}{i} \binom{k+b-i-1}{b-i}, \text{ and}$$

$$b_{\min} = \min\{1 \leq b \leq r+1 \mid C_{(b)} \leq 1\}.$$

Suppose that for $\mathbf{D} = (D_1, D_2) \leq (b_{\min}, 1) \in \mathbb{Z}_{\geq 1}^2$, it holds that

$$c_{\mathbf{D}} := \dim_{\mathbb{F}} A_{\leq \mathbf{D}} / \langle [A \cdot F]_{\leq \mathbf{D}} \rangle_{\mathbb{F}} = \begin{cases} \text{Coeff}\left(G(t_1, t_2), t_1^{D_1} t_2^{D_2}\right) & \text{if } \mathbf{D} \neq (b_{\min}, 1) \\ 1 & \text{if } \mathbf{D} = (b_{\min}, 1) \end{cases}$$

where

$$G(t_1, t_2) = \frac{1}{(1 - t_1)^{v+1}} + \sum_{b=0}^{v+1} C_{(b)} t_1^b t_2.$$

Then, Algorithm 3 with the bi-degree $(b_{min}, 1) \in \mathbb{Z}_{\geq 1}^2$ (using Gaussian elimination) outputs a Gröbner basis of the ideal $\langle F \rangle_A$ with at most

$$m(M_{(b_{min}, 1)})^w$$

multiplications and divisions, and for $\mathbf{D} \in \mathbb{Z}_{\geq 1}^2$ such that $\mathbf{D} \leq (b_{min}, 1)$, Algorithm 4 with the bi-degree $\mathbf{D}$ (using Gaussian elimination) computes solutions to $F = \mathbf{0}$ with at most

$$m(M_{\mathbf{D}})^w + 2 \cdot q^{c_D} \cdot c_D \cdot M_{\mathbf{D}} \cdot m \cdot M_{(1,1)}$$

multiplications and divisions where $w = 3$.

Remark 15 (Gaussian elimination on a Macaulay matrix). The columns of the Macaulay matrix $Mac_D(F)$ correspond to the monomials of degree at most D , and the number of columns is $\binom{n+D}{D}$. By multiplying each polynomial of degree two by all the monomials of degree at most $D-2$, we can compute rows of $Mac_D(F)$. Hence, the number of rows is $m \binom{n+D-2}{D-2}$. When we use Gaussian elimination for computing an echelon form of $Mac_D(F)$, the number of multiplications and divisions is bounded from above by

$$m \binom{n+D-2}{D-2} \binom{n+D}{D}^2 \leq m \binom{n+D}{D}^3.$$

Hence, a smaller degree D provides a lower complexity when the degrees and the number of variables are fixed. Note that this discussion depends on the complexity of specific algorithms, such as Gaussian elimination, and how those algorithms are applied in a solving process.

Remark 16. In Corollary 8, we introduced a combinatorial form for the generating function of $\{\dim_{\mathbb{F}} A_{\leq i} / \langle [A \cdot F]_{\leq i} \rangle_{\mathbb{F}}\}_{i \geq 0}$. In this paper, we call $f_1, \dots, f_m \in A$ a d -regular sequence if it holds that for any $d_0 < d$ and $i \in \{1, \dots, m-1\}$,

$$\times f_{i+1} : A_{\leq d_0 - \deg f_{i+1}} / \langle [A \cdot F_i]_{\leq d_0 - \deg f_{i+1}} \rangle_{\mathbb{F}} \rightarrow A_{\leq d_0} / \langle [A \cdot F_i]_{\leq d_0} \rangle_{\mathbb{F}}, \quad \bar{g} \mapsto \overline{f \cdot g},$$

is injective where $F_i = \{f_1, \dots, f_i\}$ (see also Appendix A and B). Suppose that $c = \dim_{\mathbb{F}} A / \langle F \rangle_A < \infty$ and $d_{min} = \min\{d \geq c \mid \dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \leq c\} < \infty$. Then we can obtain that $f_1, \dots, f_m$ is d_{min} -regular if and only if it satisfies (7) (see Appendix B). Note that, for $c = 0$, this fact has essentially already been proved in [10]. As mentioned in [9, 23], these discussions run in parallel with those presented in Appendix (see Appendix A).

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Appendix

Appendix A: Relation with Hilbert series

Following [23], we have that the generating function of $\{\dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}\}_d$ coincides with that of $\{\dim_{\mathbb{F}} S_d / \langle F^h \rangle_d\}_d$ as follows.

Standard degree case Let $S = \mathbb{F}[x_0, x_1, \dots, x_n]$. We denote by S_d the set of all homogeneous polynomials of degree d in S , i.e. $S_d = \langle \text{Mon}_d(\{x_0, x_1, \dots, x_n\}) \rangle_{\mathbb{F}}$. Then there exists a direct sum decomposition $S = \bigoplus_{d \geq 0} S_d$ as a vector space. An ideal generated by homogeneous polynomials is called a *homogeneous ideal*. For a homogeneous ideal I and a non-negative integer d , there exists a direct sum decomposition $I = \bigoplus_{d \geq 0} I_d$ where $I_d = I \cap S_d$. Then, in [8], the following generating function is called the *Hilbert series* of S/I :

$$\text{HS}_{S/I}(t) := \sum_{d \geq 0} (\dim_{\mathbb{F}} S_d / I_d) \cdot t^d.$$

Lemma 9. *Let $A = \mathbb{F}[x_1, \dots, x_n]$ and $S = \mathbb{F}[x_0, x_1, \dots, x_n]$, and let I be a homogeneous ideal of S . Then followings hold:*

1. $\dim_{\mathbb{F}} A_{\leq d} = \dim_{\mathbb{F}} S_d$
2. $\dim_{\mathbb{F}} S_d / I_d = \dim_{\mathbb{F}} S_d - \dim_{\mathbb{F}} I_d$

Definition 10. *For a polynomial $f \in A = \mathbb{F}[x_1, \dots, x_n]$, we define the homogenization $f^h \in S = \mathbb{F}[x_0, x_1, \dots, x_n]$ of f by a new variable x_0 as follows:*

$$f^h(x_0, x_1, \dots, x_n) := x_0^{\deg f} \cdot f(x_1/x_0, \dots, x_n/x_0).$$

Note that $f^h(x_0, x_1, \dots, x_n)$ is a homogeneous polynomial of degree $\deg f$ in S . Hence for polynomials $f_1, \dots, f_m$ in A , an ideal generated by $f_1^h, \dots, f_m^h$ in S is a homogeneous ideal and there exists a direct sum decomposition

$$\begin{aligned} \langle f_1^h, \dots, f_m^h \rangle &= \bigoplus_{d \geq 0} \langle f_1^h, \dots, f_m^h \rangle_d, \\ \langle f_1^h, \dots, f_m^h \rangle_d &= \{g_1 f_1^h + \dots + g_m f_m^h \mid g_i \in S_{d - \deg f_i^h}\}. \end{aligned}$$

Lemma 10 ([9,23]). *There exists an isomorphism of vector spaces as follows:*

$$\begin{aligned} \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} &= \left\langle \sum_i A_{\leq d - \deg f_i} \cdot f_i \mid 1 \leq i \leq m \right\rangle_{\mathbb{F}} \cong \langle f_1^h, \dots, f_m^h \rangle_d, \\ f(x_1, \dots, x_n) &\mapsto x_0^d \cdot f(x_1/x_0, \dots, x_n/x_0), \\ g(1, x_1, \dots, x_n) &\leftarrow g(x_0, x_1, \dots, x_n). \end{aligned}$$

Therefore we have the following corollary:

Corollary 11 ([9,23]). $\text{HS}_{\mathbb{F}[x_0, \dots, x_n] / \langle f_1^h, \dots, f_m^h \rangle}(t) = \sum_{d \geq 0} \dim_{\mathbb{F}} A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}} \cdot t^d.$

Multi-degree case For $\underline{X}_i = (x_{1i}, \dots, x_{n_i i})$, we define $\widehat{\underline{X}}_i = (x_{0i}, x_{1i}, \dots, x_{n_i i})$ and $\underline{X}_i/x_{0i} = (x_{1i}/x_{0i}, \dots, x_{n_i i}/x_{0i})$ with a new variable x_{0i} . Then, for a polynomial $f(\underline{X}_1, \dots, \underline{X}_s) \in \mathbb{F}[\underline{X}_1, \dots, \underline{X}_s]$ with $\deg_{\mathbb{Z}_{\geq 0}^s} f = (d_1, \dots, d_s)$, we define the homogenized polynomial $f^h(\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s)$ with respect to $x_{01}, \dots, x_{0s}$ as follows:

$$f^h(\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s) := x_{01}^{d_1} \cdots x_{0s}^{d_s} \cdot f(\underline{X}_1/x_{01}, \dots, \underline{X}_s/x_{0s}).$$

This is a homogeneous polynomial in $\mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s]$. For $f_1, \dots, f_m \in \mathbb{F}[\underline{X}_1, \dots, \underline{X}_s]$, the ideal generated by $f_1^h, \dots, f_m^h \in \mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s]$ is a homogeneous ideal:

$$\langle f_1^h, \dots, f_m^h \rangle = \bigoplus_{\mathbf{d} \in \mathbb{Z}_{\geq 0}^s} \langle f_1^h, \dots, f_m^h \rangle_{\mathbf{d}},$$

$$\langle f_1^h, \dots, f_m^h \rangle_{\mathbf{d}} = \{g_1 f_1^h + \cdots + g_m f_m^h \mid g_i \in \mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s]_{\mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f_i^h}\}.$$

Lemma 11. For $\mathbf{d} = (d_1, \dots, d_s) \in \mathbb{Z}_{\geq 0}^s$, there exists the following isomorphism of vector spaces:

$$\langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} = \left\langle \sum_i A_{\leq \mathbf{d} - \deg_{\mathbb{Z}_{\geq 0}^s} f_i} \cdot f_i \mid 1 \leq i \leq m \right\rangle_{\mathbb{F}} \cong \langle f_1^h, \dots, f_m^h \rangle_{\mathbf{d}},$$

$$f(\underline{X}_1, \dots, \underline{X}_s) \mapsto x_{01}^{d_1} \cdots x_{0s}^{d_s} \cdot f(\underline{X}_1/x_{01}, \dots, \underline{X}_s/x_{0s}),$$

$$g(\{1\} \parallel \underline{X}_1, \dots, \{1\} \parallel \underline{X}_s) \mapsto g(\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s),$$

where $\{1\} \parallel \underline{X}_i = (1, x_{1i}, \dots, x_{n_i i})$.

Hence the following assertion holds:

Corollary 12. $\sum_{\mathbf{d}=(d_1, \dots, d_s)} \dim \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} \cdot t_1^{d_1} \cdots t_s^{d_s} = \sum_{\mathbf{d}=(d_1, \dots, d_s)} \dim \langle f_1^h, \dots, f_m^h \rangle_{\mathbf{d}} \cdot t_1^{d_1} \cdots t_s^{d_s}.$

In [17], for a homogeneous ideal I in $\mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s]$, the multivariate Hilbert series $\text{HS}_{\mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s]/I}(t_1, \dots, t_s)$ is defined as

$$\text{HS}_{\mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s]/I}(t_1, \dots, t_s) = \sum_{\mathbf{d}=(d_1, \dots, d_s) \in \mathbb{Z}_{\geq 0}^s} \dim_{\mathbb{F}} \mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s]_{\mathbf{d}} / I_{\mathbf{d}} \cdot t_1^{d_1} \cdots t_s^{d_s}.$$

By Corollary 12, we have the assertion:

Corollary 13. $\sum_{\mathbf{d}=(d_1, \dots, d_s)} \dim A_{\leq \mathbf{d}} / \langle [A \cdot F]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} \cdot t_1^{d_1} \cdots t_s^{d_s} = \text{HS}_{\mathbb{F}[\widehat{\underline{X}}_1, \dots, \widehat{\underline{X}}_s] / \langle f_1^h, \dots, f_m^h \rangle}(t_1, \dots, t_s).$

Regularity assumption In Remark 16, we mention the regularity of a polynomial sequence and show in this section that it is equivalent to that of a sequence generated by homogenized polynomials. We know the following lemma about the commutativity of a multiplication by a polynomial and homogenization:

Lemma 12. Let $\varphi_{\mathbf{d}}$ be the isomorphism $A_{\leq \mathbf{d}} \rightarrow S_{\mathbf{d}}, f(\underline{X}_1, \dots, \underline{X}_s) \mapsto x_{01}^{d_1} \dots x_{0s}^{d_s} \cdot f(\underline{X}_1/x_{01}, \dots, \underline{X}_s/x_{0s})$ where $\mathbf{d} = (d_1, \dots, d_s)$. Then the following diagram is commutative:

$$\begin{array}{ccc}
A_{\leq \mathbf{d} - \deg f_{i+1}} / \langle [A \cdot F_i]_{\leq \mathbf{d} - \deg f_{i+1}} \rangle_{\mathbb{F}} & \xrightarrow{\times f_{i+1}} & A_{\leq \mathbf{d}} / \langle [A \cdot F_i]_{\leq \mathbf{d}} \rangle_{\mathbb{F}} \\
\downarrow \overline{\varphi}_{\mathbf{d} - \deg f_{i+1}} & \circlearrowleft & \downarrow \overline{\varphi}_{\mathbf{d}} \\
S_{\mathbf{d} - \deg f_{i+1}} / \langle F_i^h \rangle_{\mathbf{d} - \deg f_{i+1}} & \xrightarrow{\times f_{i+1}^h} & S_{\mathbf{d}} / \langle F_i^h \rangle_{\mathbf{d}}
\end{array}$$

Definition 11. Let $\mathbf{d} \in \mathbb{Z}_{\geq 0}^s$. We call $f_1, \dots, f_m$ a $\mathbf{d}$ -regular sequence if it holds that for any $\mathbf{d}_0 \preceq \mathbf{d}$ and any $i \in \{1, \dots, m-1\}$,

$$\times f_{i+1} : A_{\leq \mathbf{d} - \deg f_{i+1}} / \langle [A \cdot F_i]_{\leq \mathbf{d} - \deg f_{i+1}} \rangle_{\mathbb{F}} \rightarrow A_{\leq \mathbf{d}} / \langle [A \cdot F_i]_{\leq \mathbf{d}} \rangle_{\mathbb{F}}, g \mapsto g \cdot f_{i+1}$$

is injective.

Let $s = 1$ and $d \in \mathbb{Z}_{\geq 0}$. By Lemma 12, we can obtain that the polynomial sequence $f_1, \dots, f_m$ is d -regular sequence in Definition 11 if and only if $f_1^h, \dots, f_m^h$ is d -regular sequence in the sense of [2,10].

Appendix B: The proof in Remark 16

Let $F = \{f_1, \dots, f_m\}$, $R_d = A_{\leq d} / \langle [A \cdot F]_{\leq d} \rangle_{\mathbb{F}}$, $G_{m,n} = (1 - t^2)^m (1 - t)^{-(n+1)}$, $D_{(c)} = \min\{d \geq c \mid \text{Coeff}(G_{m,n}, t^d) \leq c\}$, and $D_{(0)} = \min\{d \geq 0 \mid \text{Coeff}(G_{m,n}, t^d) \leq 0\}$. By Appendix A, in parallel with Theorem 1 in [10], Lemma 13 below holds. If F satisfies (7), then $d_{\min} = D_{(c)}$ and F is $d_{\min}$ -regular by Lemma 13.

Lemma 13 ([10]). Let $f_1, \dots, f_m$ be quadratic polynomials in $\mathbb{F}[x_1, \dots, x_n]$. The following assertions are equivalent:

1. $f_1, \dots, f_m$ is d -regular.
2. $\forall i < d$, $\dim_{\mathbb{F}} A_{\leq i} / \langle [A \cdot F]_{\leq i} \rangle_{\mathbb{F}} = \text{Coeff}((1 - t^2)^m (1 - t)^{-(n+1)}, t^i)$

We assume that F is $d_{\min}$ -regular. Then, for any $d < d_{\min}$, it holds that $d < c$ or $\text{Coeff}(G_{m,n}, t^d) = \dim_{\mathbb{F}} R_d > c$ by definition and Lemma 13. Hence, we have $D_{(c)} \geq d_{\min}$. Note that $D_{(0)} \geq d_{\min}$ holds even when $c \geq 1$. In fact, for any $d < d_{\min}$, we have $\text{Coeff}(G_{m,n}, t^d) = \dim_{\mathbb{F}} R_d \geq 1$ by Lemma 6. Hence, $D_{(0)} \geq d_{\min}$.

Suppose $D_{(c)} > d_{\min}$, and we will derive a contradiction. Since $D_{(0)} \geq d_{\min}$, we have $\text{Coeff}([G_{m,n}]_{(0)}, t^{d_{\min}}) \geq \text{Coeff}(G_{m,n}, t^{d_{\min}})$ by definition of $[\cdot]_{(0)}$ (see Definition 9). Moreover, we have $\dim_{\mathbb{F}} R_{d_{\min}} \geq \text{Coeff}([G_{m,n}]_{(0)}, t^{d_{\min}})$ by Lemma 13 and b) of Proof of The Theorem in [15] or b) of Proposition 1 in [10]. Then, since $d_{\min} \geq c$, we have a contradiction

$$\text{Coeff}(G_{m,n}, t^{d_{\min}}) > c \geq \dim_{\mathbb{F}} R_{d_{\min}} \geq \text{Coeff}([G_{m,n}]_{(0)}, t^{d_{\min}}) \geq \text{Coeff}(G_{m,n}, t^{d_{\min}}).$$

Therefore, we obtain $D_{(c)} = d_{\min}$. By Remark 6, we have $\dim_{\mathbb{F}} R_{d'} = c$ for any $d' \geq d_{\min}$, and it follows that F satisfies (7) if it is $d_{\min}$ -regular.